

Leo Greenwood

Email: leogreenwood813@gmail.com

Telephone: 07999886133

Linkedin: www.linkedin.com/in/leo-greenwood-288032261

Portfolio: <https://leos-portfolio-7dus.onrender.com/>

2nd Year mechanical engineering student with a strong interest in mechanical design, simulation and optimisation and the automotive business. Strengths in CAD, simulation (FEA, CFD) and teamworking backed by winning the The Dr Trish Murray Professional Behaviours Team Award.

EDUCATION

University of Sheffield:

Sep 2024 - Current

MEng: Mechanical Engineering with an Industrial Placement Year

Currently achieving a 1st (77% average)

Relevant Modules: Dynamics of Structures and Machines (73%), Mechanics of Deformable Solids (62%), Design Project Part 1 (78%), Mathematics for Engineering Modelling (84%), Accounting and Law for Engineers (86%)

Barnsley Sixth form College:

Sep 2022 - June 2024

A-Levels: Mathematics (A)

A-Levels: Physics (A)

A-Levels: Geography (B)

SKILLS

- Fusion 360, AutoCAD, Ansys Fluent
- MATLAB, Python (Pandas, SciPy, Flask)
- Excel, PowerPoint, Word
- Experience with component testing, manual assembly, and mechanical hand tools
- Presenting skills, communication and teamworking experience

AWARDS

The Dr Trish Murray Professional Behaviours Team Award

ACADEMIC ENGINEERING EXPERIENCE

2nd Year Design Project:

Oct 2025 - May 2026

- Designed and manufactured a functional gearbox compliant with EAPC regulations as part of a 5 person team
- Conducted sensitivity studies by modelling components in Fusion 360 using FEA to test for failures in components such as the gear teeth
- Assembled and ran physical prototype testing to evaluate mechanical efficiency and component durability, achieving an upper second class grade

Engineering: You're Hired:

Feb 2026

- Engineered a cost effective and sustainable solution to address the growing emissions within the aviation industry
- Delivered the project within a 5 day deadline which enhanced communication, teamwork and problem solving skills
- Won the best design in the hub alongside The Dr Trish Murray Professional Behaviours Team Award

1st Year Design Project:

Feb - May 2026

- Collaborated within a 5 person team to design, manufacture and test a bottle rocket
- Developed a MATLAB script to model both thrust and projectile phases to predict rocket trajectory
- Conducted launch tests to compare flight distances against the MATLAB model

PERSONAL PROJECTS

Quarter Car Suspension Simulation:

June 2026 - Current

- Developed a 2 DOF state space simulation in Python and MATLAB to model chassis displacement and peak body acceleration of sprung and unsprung masses over an 80mm speed bump
- Implemented a loop to evaluate a range of damping ratios, identifying the optimal balance to minimise settling and peak body acceleration
- Designed a complete double wishbone suspension system including the knuckle, upper/lower wishbones and the spring damper unit in Fusion 360 utilising Sheffield Eco-Motorsport data specifications
- Plan to conduct FEA studies in Ansys Mechanical to validate structural integrity and optimise the design

VIRTUAL WORK EXPERIENCE

Bright Network's IEUK 2026:

June 2026

- Completed a virtual sector skills experience, engaging in industry panels on sustainable infrastructure and engineering workflows
- Programmed a data processing script using Python and Pandas to analyse large datasets for an AeroGrid simulation utilising docker for containerisation
- Engaged in professional networking workshops, connecting with peers and industry professionals to develop professional communication and career insights